HIP Final Project

Frontal Theta Power as a Neural Marker of Cognitive Load in Verbal Working Memory: A NeuroPype-Based EEG Analysis

Team 9

111090025 林佳誼
114233503 柯亦安
114233508 林謙孜

Outline

1. Introduction

Working memory is a core cognitive function that allows humans to temporarily store and manipulate information. It plays an essential role in everyday tasks such as following instructions, mental calculation, and language comprehension. However, working memory has a limited capacity, and when the amount of information exceeds this capacity, cognitive overload occurs. Understanding how the brain responds to increasing working memory demand is important for both cognitive neuroscience and practical applications such as workload monitoring. EEG provides a non-invasive means of measuring brain activity with high temporal resolution, making it a suitable tool for tracking real-time changes in cognitive load. One well-established EEG marker of working memory is frontal theta oscillations, rhythmic activity in the 4-8 Hz frequency range recorded over the frontal cortex. Previous research has shown that frontal theta power increases with increasing working memory load, reflecting the greater neural effort required to maintain and process information in the prefrontal cortex.

In this project, we investigate how frontal theta power changes across three levels of cognitive load using a digit-span task, in which participants memorize sequences of 5, 9, or 13 digits. A control condition in which participants listened to the digit sequences without memorizing them was also included for comparison. EEG data were processed using NeuroPype to extract frontal theta power at electrode Fz across all conditions.

We hypothesize that theta power will be significantly higher in the memory condition compared to the control condition, and that theta power will increase systematically as the number of digits increases from 5 to 13. If supported, these findings would confirm that frontal theta power is a reliable and sensitive neural indicator of verbal working memory load.

2. Data Description

<u>Data Source & Recording Info.</u>

- The EEG data analyzed in this project was acquired from the open-access OpenNeuro database.
- We utilized dataset ds003838 (Version 1.0.6) and specifically extracted the data from Subject 40 (Sub-040) for our end-to-end pipeline.
- The raw EEG signals were recorded utilizing a 63-channel EEG system configuration.

<u>Task and Procedure</u>

- This dataset was originally collected to investigate the neural mechanisms and cognitive load associated with verbal working memory.
- The experimental paradigm consisted of a digit-span task. As illustrated in the experimental design, each trial begins with a start screen (0.5s), followed by an instruction cue (1s) directing the participant to either "Memorize" (active memory task) or "Just listen" (control condition).
- After a 3s baseline period, participants were presented with an auditory sequence of 5, 9, or 13 digits (2s per digit, and 3s for the final digit).
- Following the auditory sequence, participants entered a serial recall phase (lasting 7, 11, or 15s, depending on the sequence length), concluding with a 5s inter-trial interval (ITI).

<u>Event Markers Structure (Event Codes)</u>

- To accurately align our EEG pipeline with the experimental events, we utilized the event codes embedded in the physiological recordings. These codes mark the presentation of each digit and contain specific experimental information.
- The event codes consist of a multi-digit sequence (e.g., 5001131), formatted as follows:
    - **Task (Digits 1-2):** 50indicates "just listen" (control); 60 indicates "memorize".
    - **Load (Digits 3-4):** Ranges from 01 to 13, indicating the specific position of the digit currently being presented in the sequence.
    - **Sequence Length (Digits 5-6):** 05, 09, or 13, representing the total number of digits in the current trial.
    - **Recall (Digit 7):** Present only in the memorize task. 0 indicates the sequence was forgotten; 1 indicates it was successfully recalled. (Note: Control trials "just listen" lack this 7th digit, resulting in a 6-digit code).

<u>Event Markers & Segmentation Rules</u>

- To extract the EEG signals corresponding to different cognitive load levels, we time-locked our data to the onset of the **first digit** in the sequence (Load = 01).

- The duration of the data segment extracted is directly proportional to the length of the digit sequence (approximately double the number of digits in seconds) to fully capture the memorization process.
- **For the 5-digit condition:** Trials were identified using the markers ['500105', '6001050', '6001051'] (representing the 1st digit of a 5-digit sequence across control, memory-forgotten, and memory-recalled conditions). The data was segmented from the marker onset to 10 seconds post-marker ([0, 10]).
- **For the 9-digit condition:** Trials were identified using the markers ['500109', '6001090', '6001091'] (representing the 1st digit of a 9-digit sequence). The data was segmented for 18 seconds ([0, 18]).
- **For the 13-digit condition:** Trials were identified using the markers ['500113', '6001130', '6001131'] (representing the 1st digit of a 13-digit sequence). The data was segmented for 26 seconds ([0, 26]).

3. Data Preprocessing

Step 1 :

Download .set file from the link below, we selected subject number 40 as our data source.

https://openneuro.org/datasets/ds003838/versions/1.0.6

Step2 :

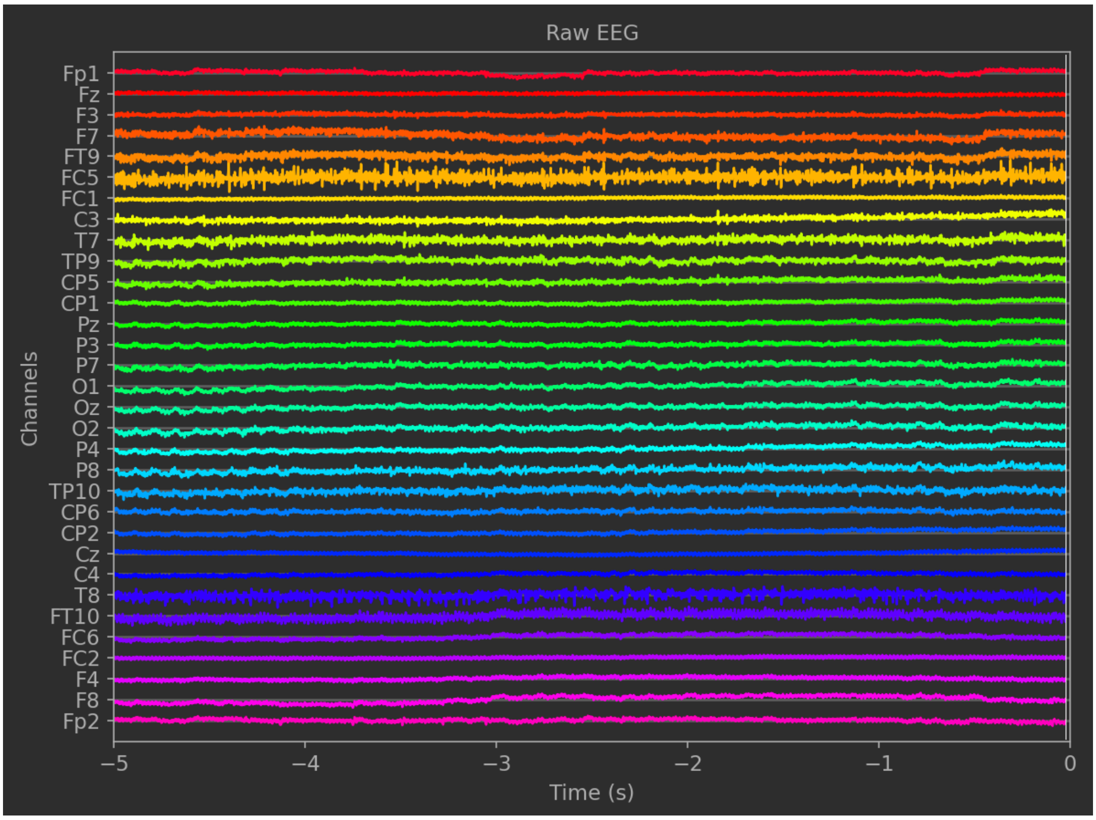
Use MATLAB to convert data into a format that can be read by a pipeline.

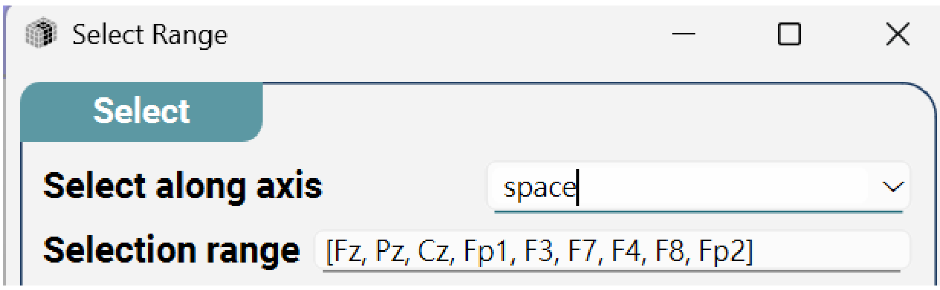
Run *convert.m* first, then run *fix_np_set_format.m*.

convert.m and *fix_np_set_format.m* will be available on GitHub.

Step3 :

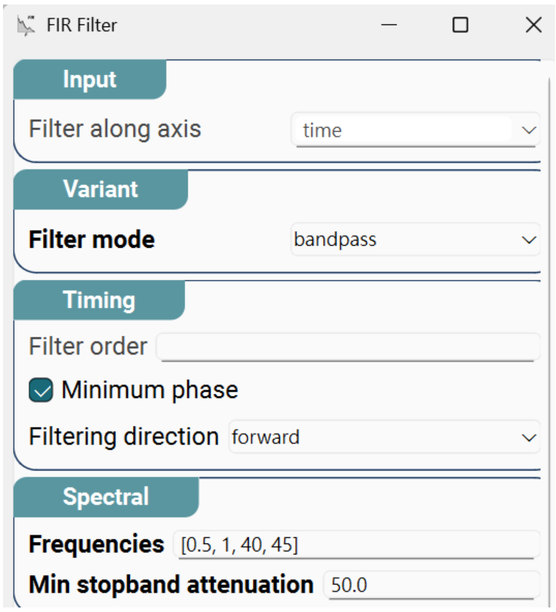
This is a time-domain plot of 63 channels.

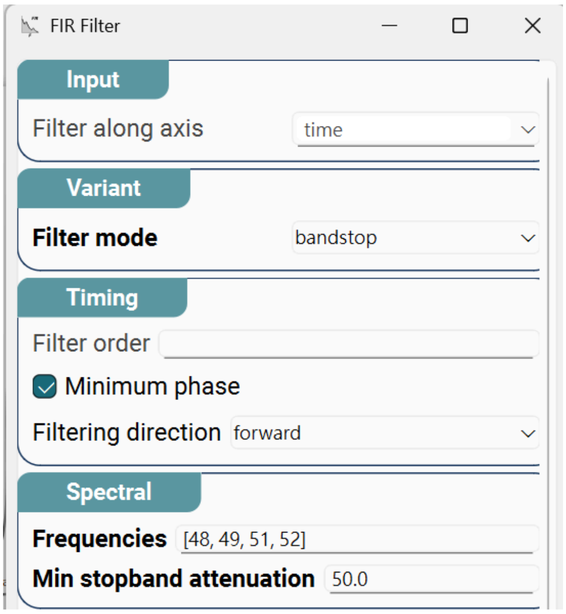




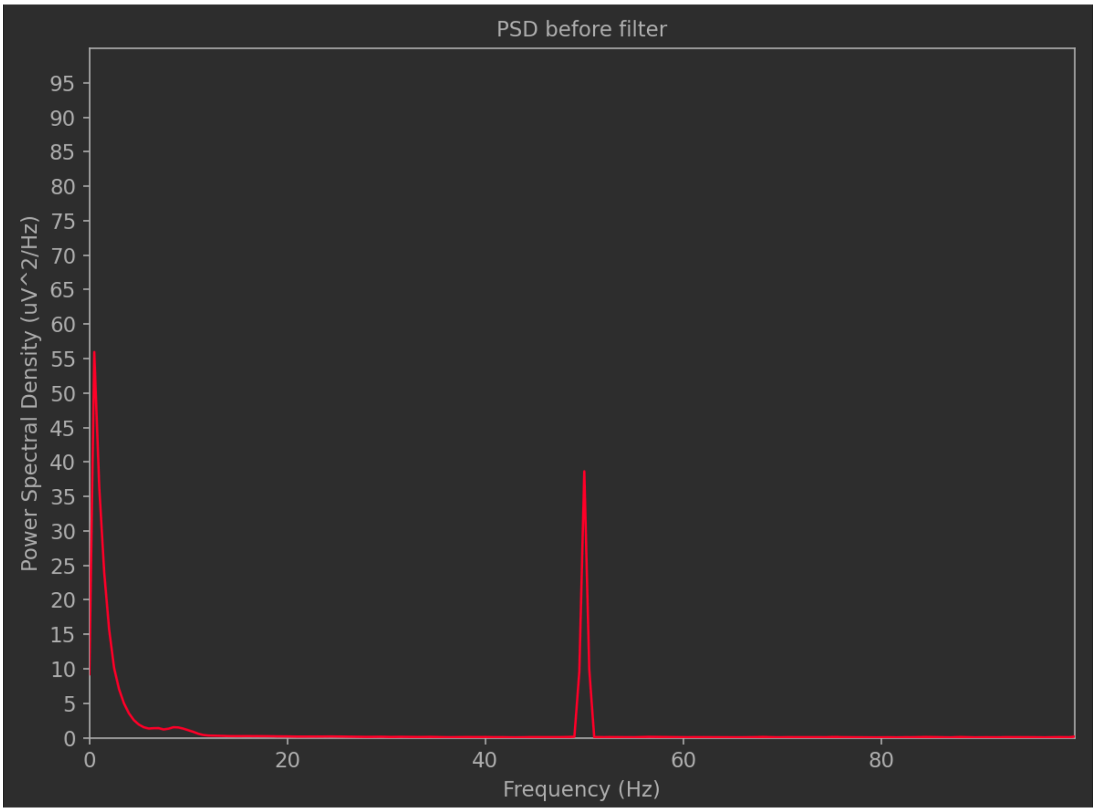
Use the **select range** function to extract the channels from the 63 channels until only channels Fz, Pz, Cz, Fp1, F3, F7, F4, F8, and Fp2 remain.

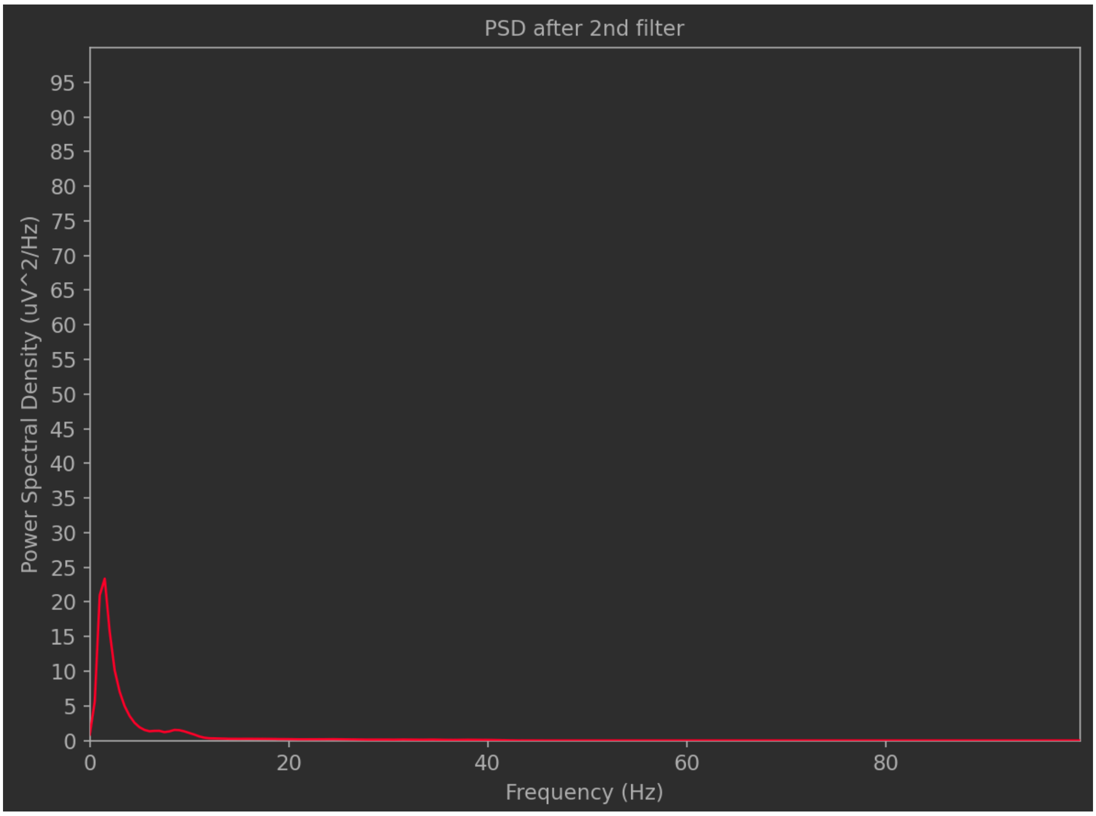
Then, an FIR filter was used to remove the 50 Hz power supply noise.
You can see the results of before and after in the PSD below.



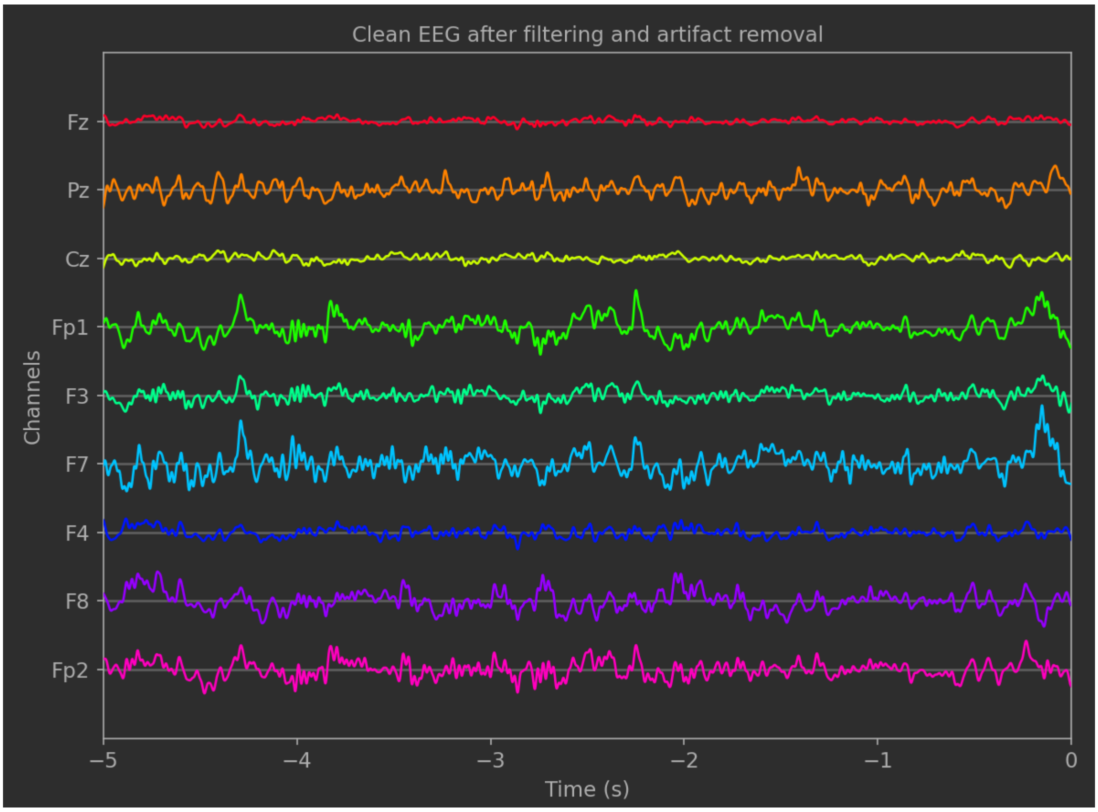


The first filter allows frequencies from 1 to 40 Hz to pass through, and the second filter removes frequencies up to 50 Hz.



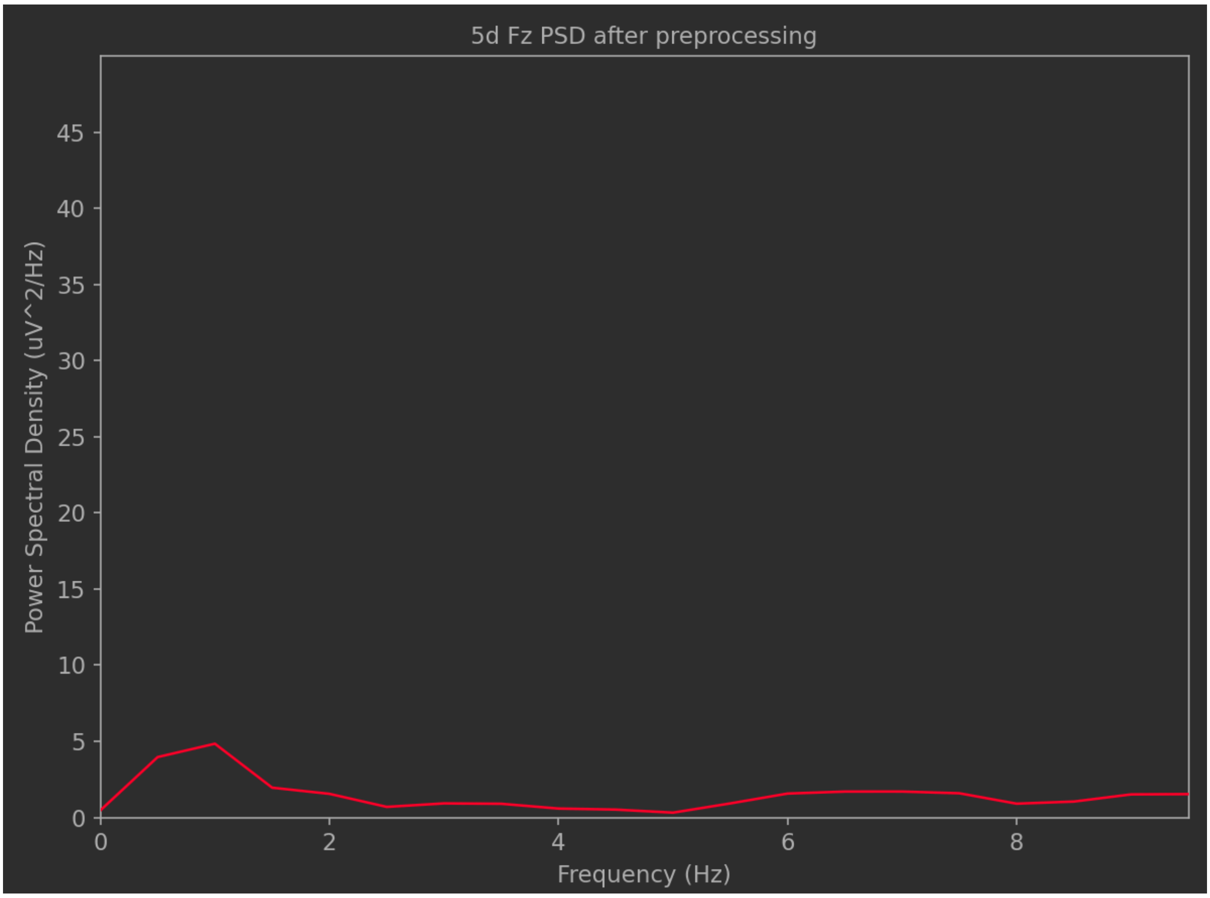


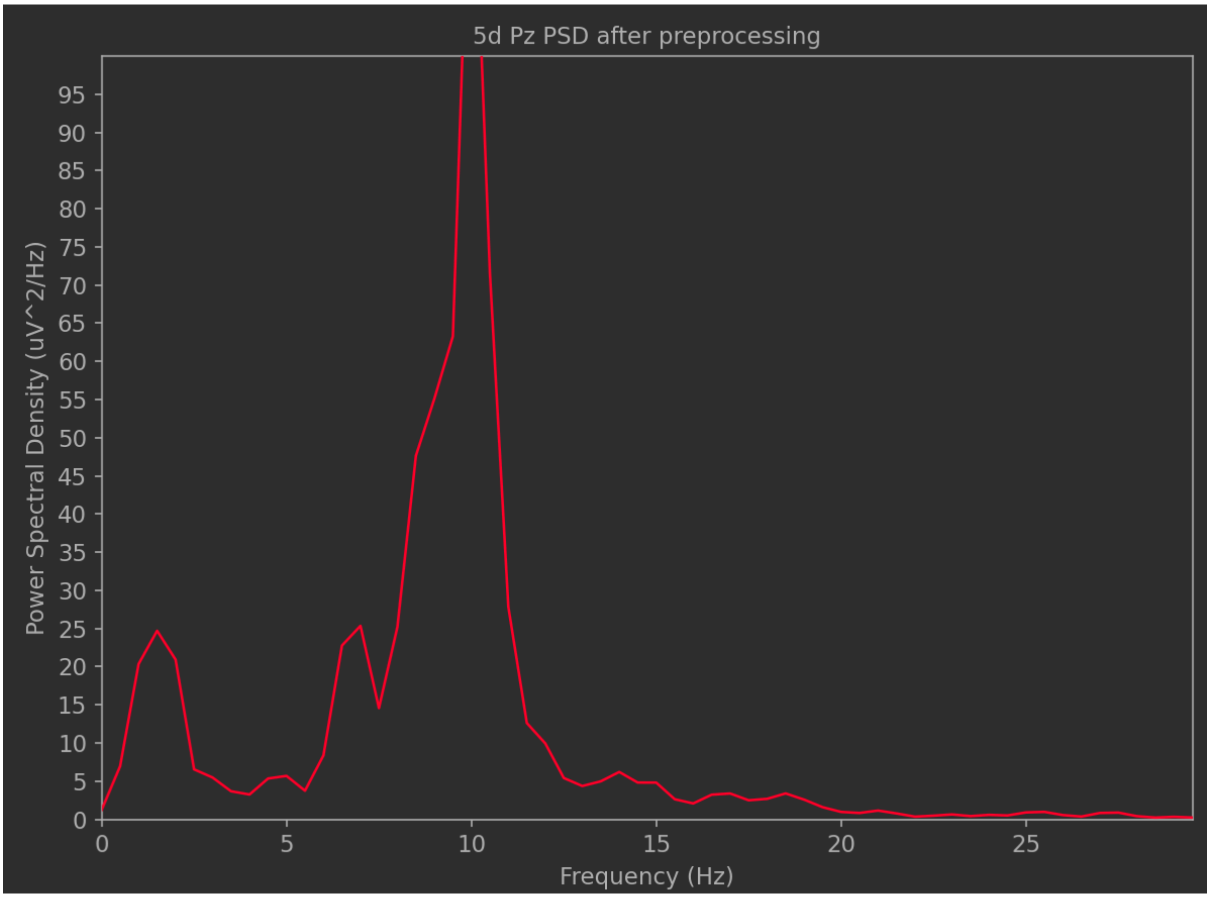
This is the time-domain plot after AR processing.



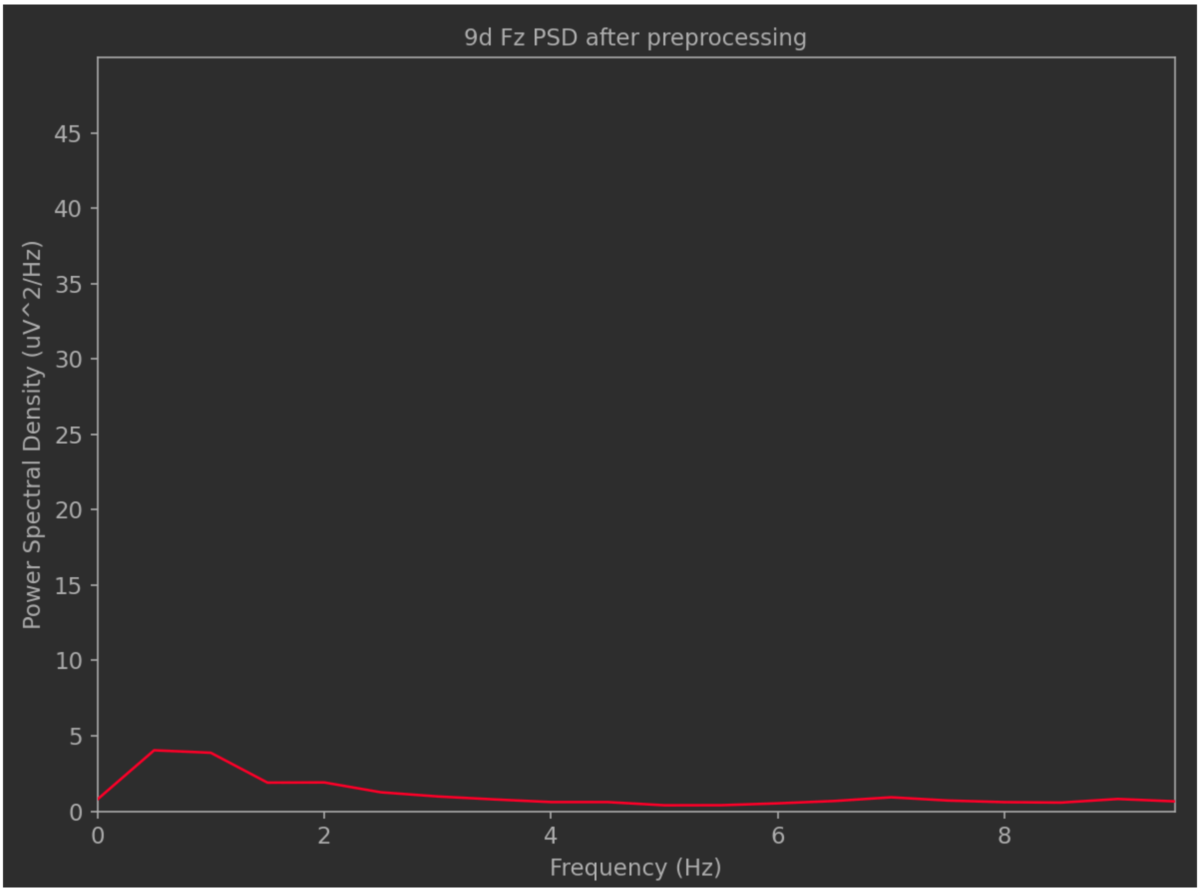
The following are PSDs with 5, 9, or 13 digits in Fz and Pz respectively.

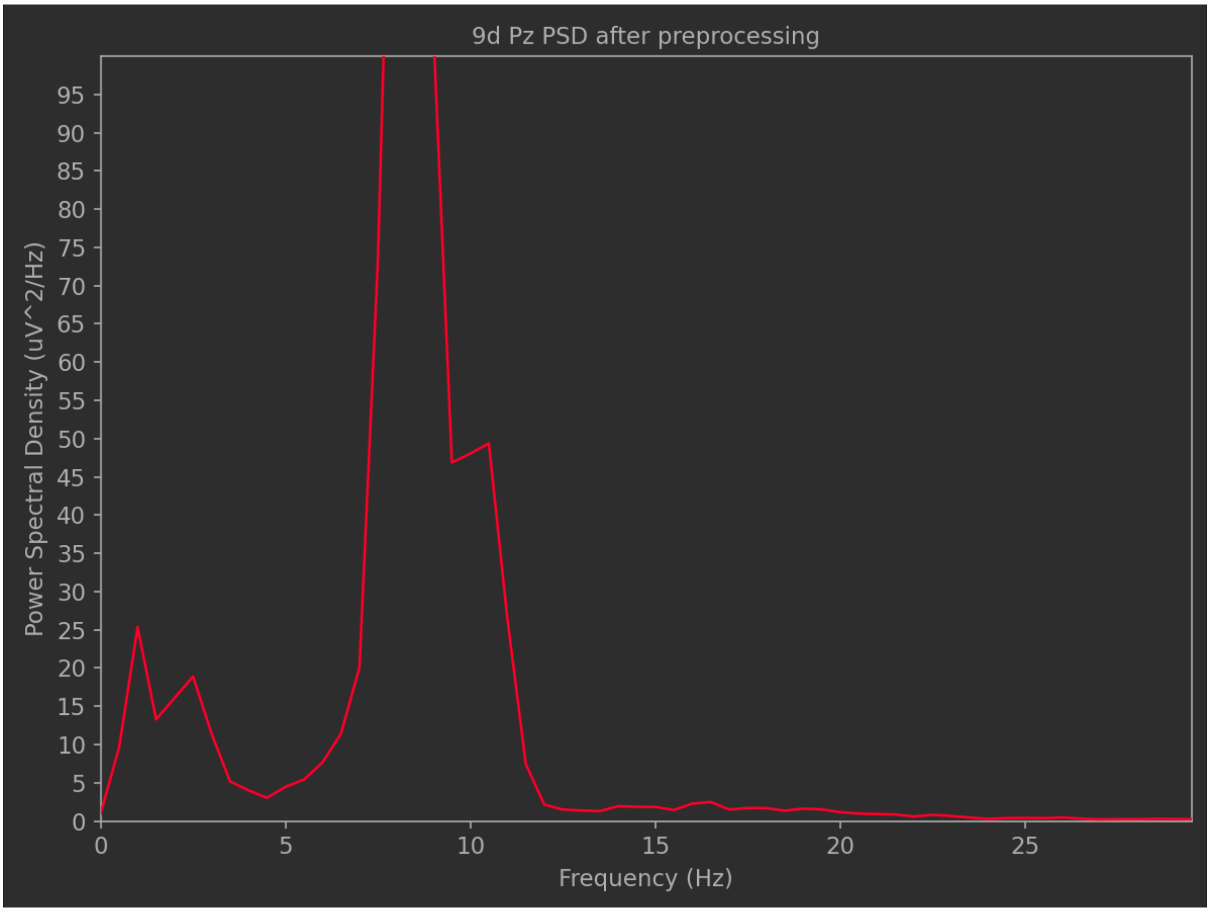
5d


5d Fz PSD after preprocessing
Power Spectral Density (uV^2/Hz)
Frequency (Hz)

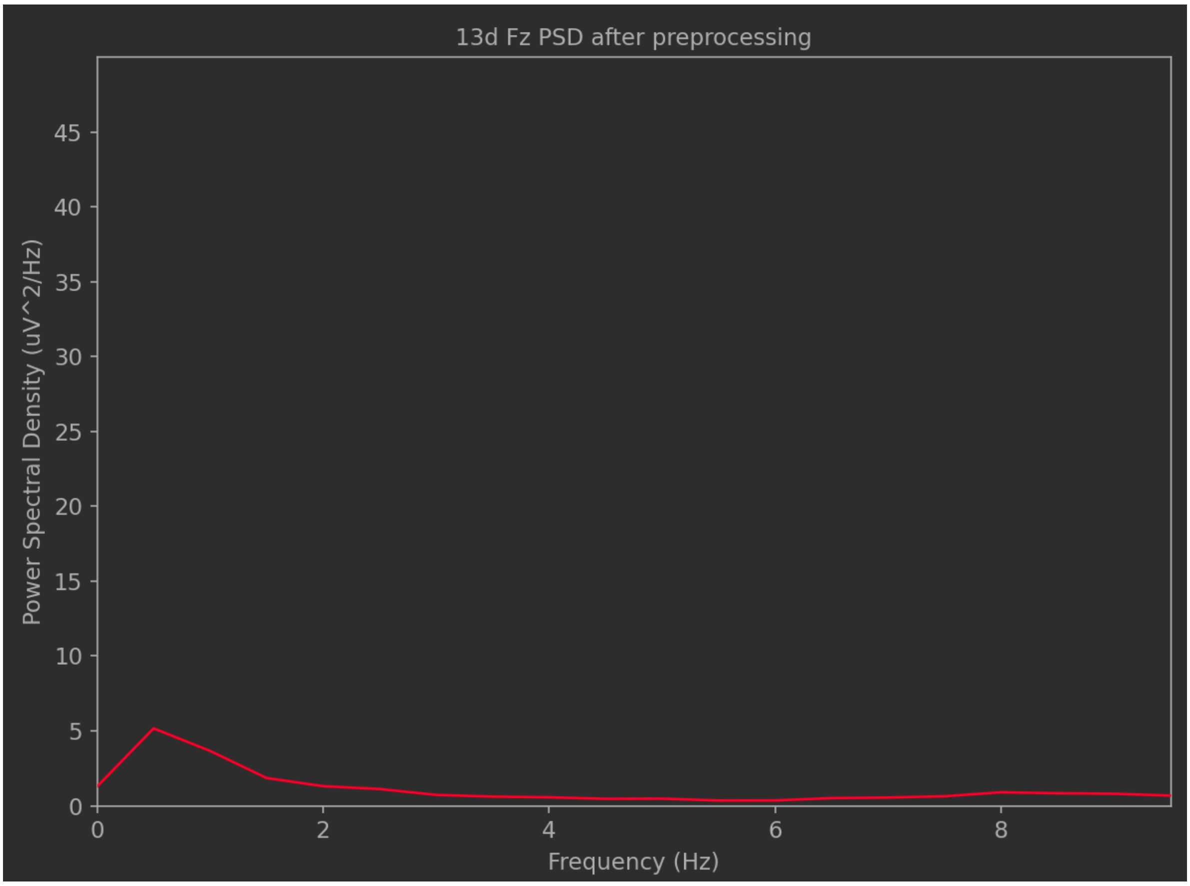

5d Pz PSD after preprocessing
Power Spectral Density (uV^2/Hz)
Frequency (Hz)

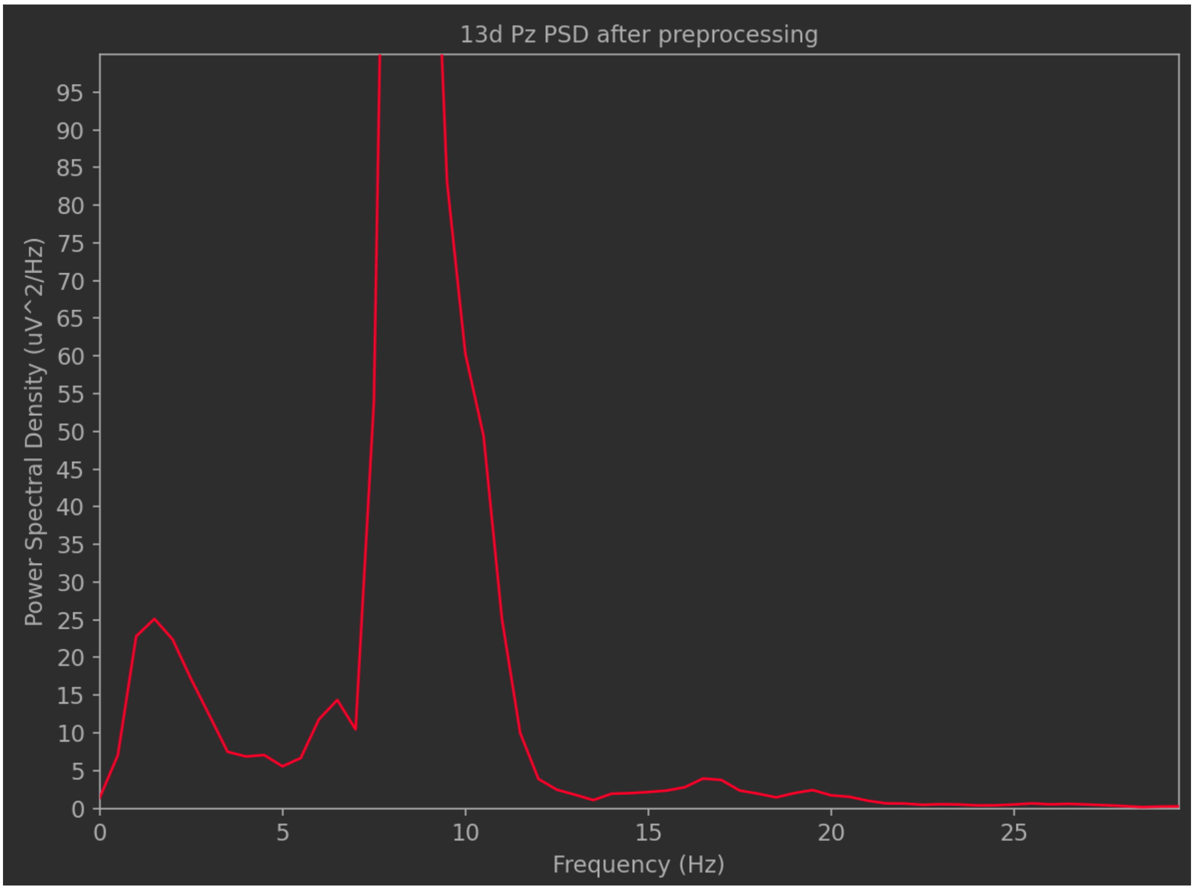
9d


9d Fz PSD after preprocessing
Power Spectral Density (uV^2/Hz)
Frequency (Hz)
0
5
10
15
20
25
30
35
40
45
0
2
4
6
8

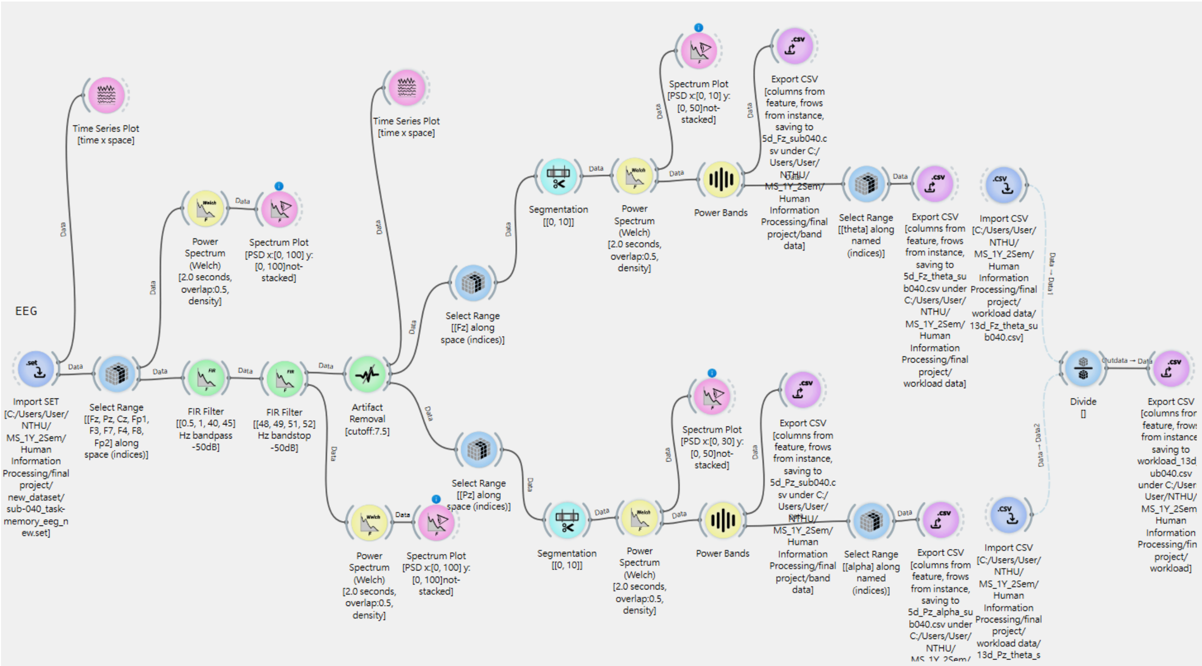

9d Pz PSD after preprocessing
Power Spectral Density (uV^2/Hz)
Frequency (Hz)
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
95
0
5
10
15
20
25

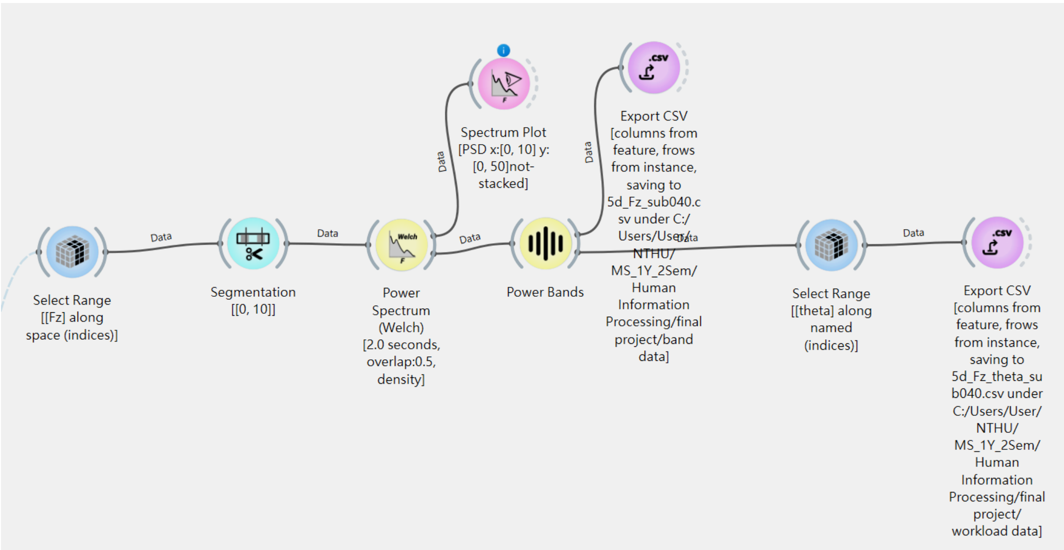
13d

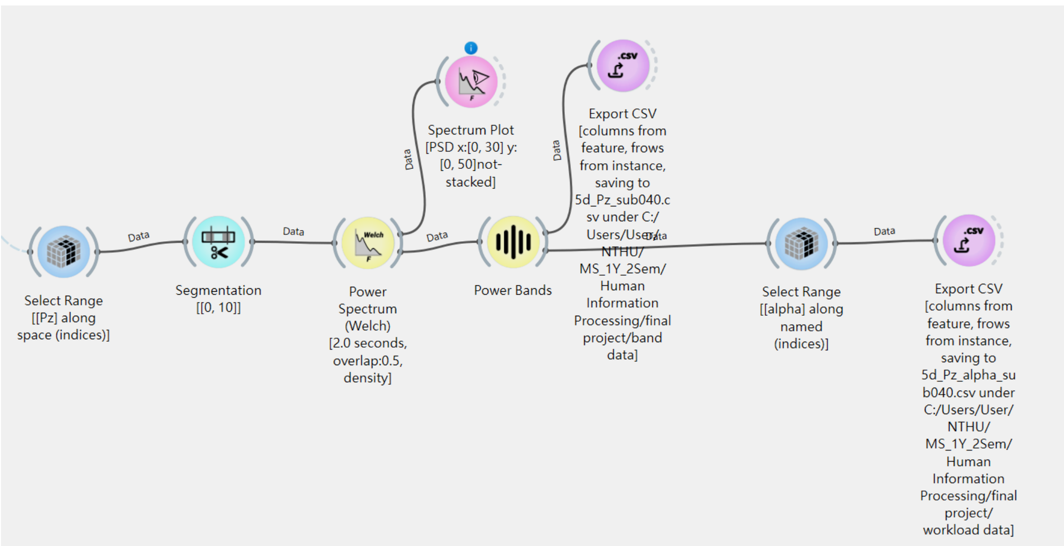

13d Fz PSD after preprocessing
Power Spectral Density (uV^2/Hz)
45
40
35
30
25
20
15
10
5
0
0
2
4
6
8
Frequency (Hz)


13d Pz PSD after preprocessing
Power Spectral Density (uV^2/Hz)
95
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
15
10
5
0
0
5
10
15
20
25
Frequency (Hz)

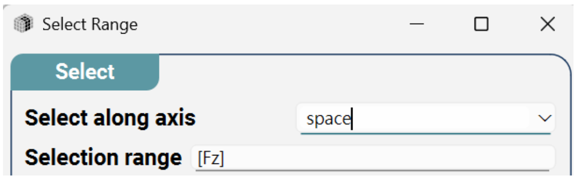
4. NeuroPype Pipeline

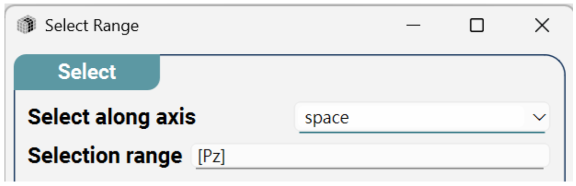




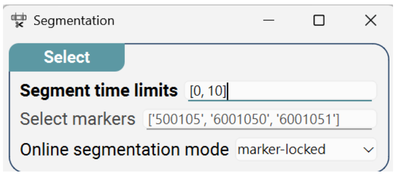


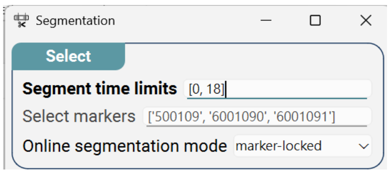
After data preprocessing, to calculate the workload, we need the theta of Fz and the alpha of Pz.
Here, we use the `**select range**` node to select the required channels

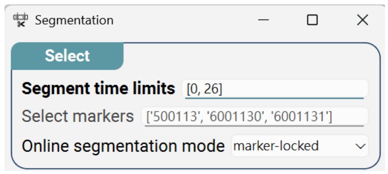




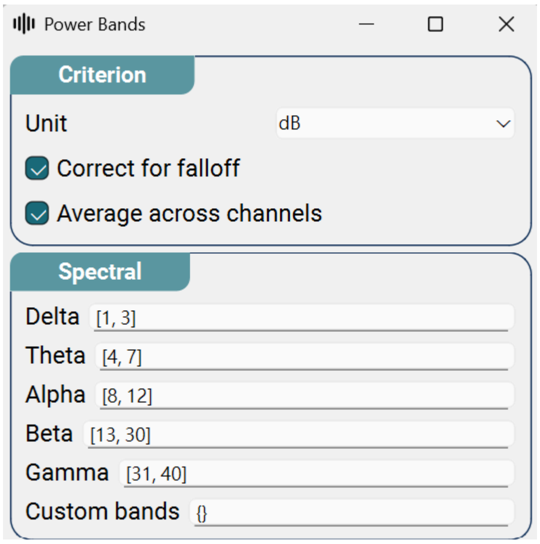
and the `**segmentation**` node to cut the data to the required length. The starting point is determined by a <u>marker</u>, and the number of seconds for segmentation varies depending on the length of the numbers: 5 digits start 10 seconds after the marker; 9 digits start 18 seconds after the marker; and 13 digits start 26 seconds after the marker. (Because the duration is different, the parameters will be changed three times.)



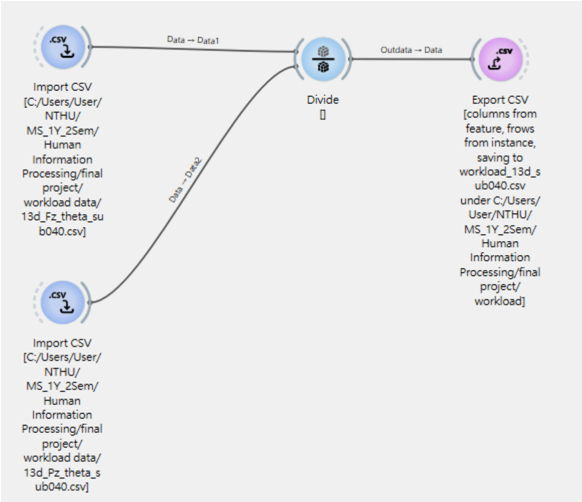




Next, we use the `**PowerBands**` node to extract features.



Finally, we use the `**select range**` node to select the theta of Fz and the alpha of Pz and output them as a CSV file.



The workload is calculated using the output CSV file, and the final output will also be a CSV file.

5. Demo Video

https://drive.google.com/file/d/1LdEuGH4EmAC9JFdJz_MTq0FhD6Vym0bD/view?usp=sharing

6. Results & Interpretation

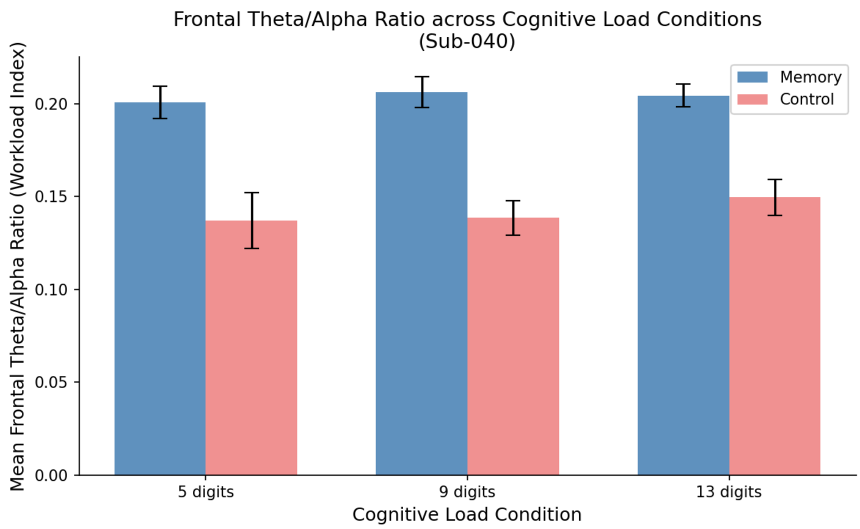


Figure 1: Frontal Theta Power across Cognitive Load Conditions

Figure 1 shows that mean frontal theta power was consistently higher in the Memory condition (~0.33) compared to the Control condition (~0.23–0.26) across all three digit-span levels for Sub-040. The difference between conditions remained stable across load levels, with no clear increase from 5 to 9 to 13 digits.

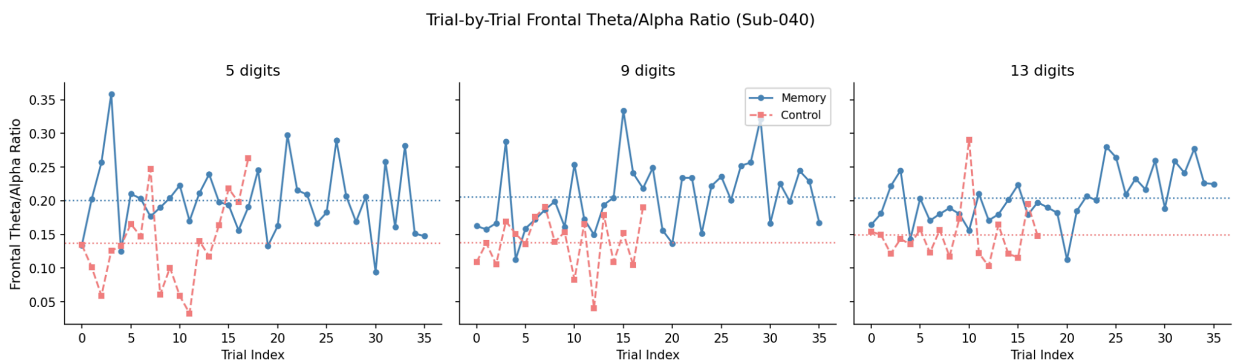


Figure 2: Trial-by-Trial Frontal Theta Power

Figure 2 confirms this pattern at the trial level. The Memory condition shows greater moment-to-moment variability and sits above the Control condition throughout all three conditions, as indicated by the dotted mean lines.

The results partially support our hypothesis. The consistently higher frontal theta power in the Memory condition compared to the Control condition across all three digit lengths aligns well with established literature on working memory and theta oscillations. When participants were required to actively encode and maintain digit sequences in working memory, the prefrontal cortex engaged more strongly, producing elevated theta-band activity at Fz. This is consistent with the prefrontal cortex's role in executive control and in maintaining information in working memory buffers. However, the expectation that theta power would increase systematically from 5 to 9 to 13 digits was not clearly observed for this participant. The mean theta power in the Memory condition remained approximately flat across all three load levels, rather than gradually increasing. One possible explanation is that Sub-040 may have reached a performance ceiling early, meaning that even the 5-digit condition was already sufficiently demanding to recruit maximum frontal theta engagement. Alternatively, individual differences in working memory capacity and strategy use could account for the relatively stable theta values across conditions.

The high trial-by-trial variability in the Memory condition may also reflect fluctuations in attentional engagement or encoding strategy across individual trials, which is a commonly reported feature in EEG working memory studies. The finding that frontal theta power reliably distinguishes between active memory encoding and passive listening provides supportive evidence that theta oscillations are a valid neural marker of verbal working memory load, even if a clear load-dependent gradient was not observed in this single-subject analysis.